Keywords for Self-Reflection (Continued)

introspect / look inward / self-examine
I might be wrong / I could be mistaken / I may be off
I could be in error / I might be misreading this / I may be overlooking a detail
I'm not sure / I'm uncertain / I'm not certain
I'm not fully confident / I can't say with certainty / I have doubts
confidence / certainty / assurance
credence / confidence level / subjective probability
revise / adjust / update
modify / refine / rework
reconsider / rethink / take another look
check my assumptions / test my assumptions / validate my premises
audit my premises / question my starting points / recheck my presuppositions
self-critique / self-review / self-audit
self-correction / self-check / critical reflection
let me check / let me verify / let me double-check
alternatively / as an alternative / as another option
instead / in lieu of that / in place of that
conversely / on the flip side / the other way around
I'm struggling with / I'm wrestling with / I'm having a hard time with

The keywords and corresponding replacements for the Self-Exploration bond are listed in the box below:

Keywords for Self-Exploration

maybe / perhaps / might be
now / at this point / right now
let's / let us / we can
probably / likely / presumably
seems / appears / looks like
maybe not / perhaps not / possibly note
I'll / we'll / it helps to
consider/ think about / look at
assume / suppose / let's say
if / provided that / in case
explore / look into / examine
probe / dig into / unpack
consider two cases / split into two cases / handle two scenarios
self-exploration / personal exploration / inner exploration
self-discovery / discovering myself / learning about myself
values / principles / priorities
I want / I'd like / I'm aiming to

Both datasets preserve the same underlying trajectories and labels as the original SFT corpus. We further ensure that the distributions of problems, answer types, and trajectory lengths remain unchanged.

D Semantic Isomer Construction Details

This section details the construction and analysis of semantic isomers across four key dimensions. We first describe the distillation of well-structured semantic isomers and their simulation via in-context learning (ICL). Subsequently, we examine information flow through metacognitive oscillation analysis and conclude with the specifics of conflict learning between two stable structures.

Model	GSM8K	MATH-500	AIME2024	AIME2025	AMC2023	OlympiadBench	AVG
LLaMA-3.1-8B-Base	7.58	3.20	0.00	0.00	4.22	1.19	2.70
+ 20K R1-Distill-Data	63.38	30.60	0.21	0.42	14.22	8.30	19.52
+ 20K OSS-Distill-Data	75.89	54.20	4.38	6.46	37.34	23.85	33.69
+ 20K QwQ-Distill-Data	64.53	32.20	2.92	0.42	16.72	8.89	20.95
Llama-3.1-8B-Instruct	75.89	35.20	4.17	1.04	23.59	12.00	25.32
+ 20K R1-Distill-Data	79.91	60.60	2.50	3.96	33.13	23.85	33.99
+ 20K OSS-Distill-Data	79.00	60.80	10.83	7.71	47.03	30.22	39.27
+ 20K QwQ-Distill-Data	82.41	60.80	4.38	8.33	32.97	25.48	35.73
Qwen-2.5-7B-Base	40.18	34.20	5.42	0.83	26.72	17.33	20.78
+ 20K R1-Distill-Data	76.14	24.20	1.20	2.29	10.00	5.33	19.86
+ 20K OSS-Distill-Data	84.99	68.40	6.04	8.13	46.25	27.70	40.25
+ 20K QwQ-Distill-Data	78.39	46.80	2.71	1.46	9.84	5.93	24.19
Qwen-2.5-7B-Instruct	83.24	74.00	12.50	7.08	22.66	38.07	39.59
+ 20K R1-Distill-Data	87.04	74.80	14.17	8.54	46.25	41.48	45.38
+ 20K OSS-Distill-Data	82.31	74.60	25.42	19.38	54.34	37.63	48.94
+ 20K QwQ-Distill-Data	85.75	73.80	13.96	16.25	52.97	35.11	46.31
Qwen-2.5-32B-Base	53.68	33.40	9.17	2.29	35.63	15.85	25.00
+ 20K R1-Distill-Data	76.14	24.20	1.46	2.29	10.00	5.33	19.90
+ 20K OSS-Distill-Data	89.76	77.20	19.38	17.50	60.63	39.26	50.62
+ 20K QwQ-Distill-Data	91.59	82.40	19.75	19.62	72.37	41.33	54.51
Qwen-2.5-32B-Instruct	93.71	81.00	15.63	14.17	69.84	42.22	52.76
+ 20K R1-Distill-Data	93.63	83.60	26.67	22.71	72.56	45.93	57.52
+ 20K OSS-Distill-Data	86.35	86.20	64.17	53.54	92.34	55.70	73.05
+ 20K QwQ-Distill-Data	92.65	93.20	59.38	49.38	88.44	61.93	74.16
LLama-3.1-70B-Base	46.78	31.80	3.33	1.88	32.50	13.19	21.58
+ 20K R1-Distill-Data	73.62	20.40	1.25	2.08	8.75	4.44	18.42
+ 20K OSS-Distill-Data	89.39	76.80	16.88	17.29	57.97	36.74	49.18
+ 20K QwQ-Distill-Data	88.86	82.40	19.58	18.33	72.34	39.41	53.49
LLama-3.1-70B-Instruct	84.23	52.20	17.92	3.13	45.63	21.63	37.45
+ 20K R1-Distill-Data	94.62	80.60	27.29	21.88	64.84	43.11	55.39
+ 20K OSS-Distill-Data	85.75	83.40	52.50	40.42	84.38	50.81	66.21
+ 20K QwQ-Distill-Data	93.33	89.00	47.50	36.25	81.41	54.07	66.93

Table 5 Full results on GSM8K, MATH-500, AIME2024, AIME2025, AMC2023, and OlympiadBench.

D.1 Distillation of Well-structured Semantic Isomers

The setting is the same as in Appendix B. We use 8 base and instruct LLMs as the backbone and distill from 3 advanced reasoning LLMs.

D.2 ICL Simulation of Semantic Isomer Structures

Inspired by Dong et al. [39], we study whether demonstration selection in an in-context learning (ICL) setting can approximate the effective semantic isomer structure of a target teacher model (Qwen2.5-32B).

Demonstration Construction. We built a candidate pool by using QwQ-32B to generate Long-CoT solutions for the training questions in Appendix B.

Demonstration Selection. For each target question, we select a 1-shot demonstration by comparing the reasoning-key distribution of each candidate trace with that of the target teacher (Qwen2.5-32B) trace for the same question, using Pearson correlation. We consider three strategies: (1) *Random:* Sample a demonstration uniformly from the pool. (2) *High-correlation (Aligned):* Choose a demonstration with high correlation ($r \gtrsim 0.9$) to the teacher distribution. (3) *Low-correlation (Mismatched):* Choose a demonstration with low correlation ($r < 0.8$), corresponding to structurally incompatible reasoning paths.

ICL-based Distillation Using the selected 1-shot demonstration, we prompt Qwen2.5-32B-Instruct to generate Long-CoT solutions for the same training questions (Appendix B). We then fine-tune a student model (e.g.,

Llama3.1-8B-Instruct) on these ICL-generated traces with the same hyperparameters as in Appendix B. For each question, we keep the demonstration fixed across all samples.

D.3 Information Flow and Metacognitive Oscillation Analysis

D.3.1 Setting: Information Flow Analysis and Metacognitive Oscillation Quantification

Information Flow Analysis in Phase Space (Human vs. R1) We compare the reasoning dynamics of humans and the R1 model in an information phase space by quantifying information gain, entropy evolution, and the rate of convergence. We use multi-step logical deduction and structured reasoning tasks from Du et al. [7]. For human data, participants wrote their reasoning process step-by-step. For model data, R1 was prompted to produce step-by-step reasoning. We treat each step as a discrete point along a reasoning trajectory. For humans, steps are segmented using natural paragraph breaks or explicit logical transitions identified by the delimiter "$\backslash n \backslash n$". For the model, steps are delineated using explicit reasoning markers in the generated output. For visualization, we report only three representative patterns of information-flow change.

Each reasoning step s_t is mapped to a semantic probability representation p_t using a unified semantic probability encoder (Llama-3.1-8B-Instruct). We define step-wise entropy through the tuple $(I_t, \Delta I_t)$, where I_t denotes the cumulative entropy (or cumulative information measure) up to step t, and $\Delta I_t = I_t - I_{t-1}$ denotes the instantaneous change. The full reasoning chain thus forms a trajectory in a 2D phase space with coordinates $(I_t, \Delta I_t)$.

To characterize reasoning dynamics (e.g., uniform gain versus accelerating convergence), we analyze the distribution of ΔI_t and the local phase-space slope, computed between consecutive points as:

$$m_t = \frac{\Delta I_t - \Delta I_{t-1}}{I_t - I_{t-1}}. \tag{47}$$

when $I_t \neq I_{t-1}$. The slope m_t captures the rate of change in information gain relative to cumulative information, indicating whether the reasoning process is accelerating, decelerating, or stable at each step.

Analysis of Metacognitive Oscillation We formalize and quantify metacognitive oscillation in LLMs—alternation between high-entropy exploration and low-entropy validation—by relating these dynamic states to the distributions of reasoning bonds. Using the phase-space dynamics, we classify each reasoning step into one of two states. (1) **High-entropy Exploration:** characterized by a steep phase-space slope ($m_t > 0.6$) together with a substantial entropy increase ($\Delta\text{entropy} > 0.05$). (2) **Low-entropy Validation:** characterized by a near-zero or negative slope ($m_t \approx 0$) and and minimal entropy change ($|\Delta\text{entropy}| < 0.05$). We then measure the frequency and periodicity of transitions between these states, and analyze which reasoning keys (bonds) are most prevalent within each state.

D.4 Details about Conflict Learning Between Two Stable Structures

Setup about performance analysis on different training data mixture strategies. To test whether simultaneously training a model on two highly correlated ($r \approx 0.9$) yet structurally distinct reasoning frameworks (from R1 and OSS) leads to "structural chaos." Specifically, we use OpenAI-OSS-120B [33] as the OSS model and DeepSeek-R1-671B-0528 [8] as the R1 model to generate two sets of Long CoT traces on the same 20K training questions from OpenThoughts-3 [24]. Based on these generated Long CoT traces, we created three training configurations: (1) OSS-Distill-Data, an R1-only set with 20K samples generated from DeepSeek-R1, (2) R1-Distill-Data, an OSS-only set with 20K samples generated from OpenAI-OSS, and (3) R1-then-OSS, a sequential set with first 10K from DeepSeek-R1 and then 10K from OpenAI-OSS. (4) OSS-then-R1, a sequential set with first 10K from OpenAI-OSS and then 10K from DeepSeek-R1. (5) R1-mix-OSS, a randomly mixed set with 10K from OpenAI-OSS and 10K from DeepSeek-R1.

Pearson correlation coefficient between transfer distribution. To quantify the similarity between two reasoning structures (e.g., R1 and OSS), we compute the Pearson correlation coefficient r between their transfer distributions. Given two transfer matrices P and Q representing the normalized frequencies of